COMPUTER VISION

Feature Tracking and Dynamic Object Detection

Mid-course project - April 2026

Objective

Detecting and localizing moving objects is an important task in many real-world applications. In surveillance and monitoring, for example, it supports early warning and event detection. Similar needs arise in traffic analysis, wildlife monitoring, and human-robot interaction, where separating moving objects from the background is a key first step for tracking and scene understanding.

The objective of this project is to analyze an input image sequence extracted from a video to detect and localize dynamic elements in the scene, providing in output a 2D bounding box. For simplicity, the localization of the detected dynamic object shall be done only in the first frame of the sequence, as shown in Figure 1. Five object categories are considered: **bird**, **car**, **frog**, **sheep**, and **squirrel**.

Figure 1: Example of selected frames from the dataset (category: frog). On the left, the first frame with a bounding box highlighting the detected object. In the center and on the right, additional frames from the sequence used as input to the system.

The system inputs and outputs are as follows:
Input:

- a set of subsequent RGB frames of a dynamic scene.

Output:

- the bounding box coordinates of each detected dynamic object in the first frame of the sequence, saved to a text file in the format detailed in the next section;
- an image showing the predicted bounding box superimposed on the first frame of the sequence.

The system should exploit temporal changes in local image features across the sequence to identify those belonging to a moving object and use them to estimate its position. Therefore, you are encouraged to design a solution based on sparse local features and to analyze their temporal association across frames through robust feature matching or optical flow techniques.

Dataset Description and Performance Measurement

The dataset to be used for the development of the project is available at the following link:

https://drive.google.com/drive/folders/18XUGoyrLiUUKTXKz1Muq7O8tn9CX4cBr?usp=sharing

The dataset[1] contains the following folders:

- **data**, whose subfolders (e.g., bird, car, etc.) contain the image sequences on which the developed system must be tested;
- **labels** (ground truth), containing bounding box annotations for the first frame of each sequence, describing the position of the object to be localized.

Annotations are provided as a text file (.txt), having one line for each category. The line is a string including the coordinates of the bounding box vertices in the format:

```
<xmin> <ymin> <xmax> <ymax>
```

The evaluation images contain different dynamic elements observed from various viewpoints and under different conditions, including illumination changes and occlusions. The system must therefore be robust to these factors, and its robustness to such variations should be analyzed in the final report.

To assess the performance of your object detection pipeline, the following metrics should be computed using the ground truth information provided in the dataset: **mean Intersection over Union** (mIoU)[2] and **detection accuracy**. Such metrics shall be used to evaluate the system performance as follows:

- **mean Intersection over Union (mIoU):** the average IoU computed across the five categories (bird, car, frog, sheep, and squirrel);
- **detection accuracy:** the number of correctly detected dynamic elements across the categories, where a detection is considered a true positive if the IoU between the predicted and the ground-truth bounding box is greater than 0.5.

Project delivery

The project must be developed in C++ with the OpenCV library **only**, and it must include a **CMakeLists.txt** file to allow compilation. **The project cannot be developed using deep learning**.

You need to deliver your project including:

- All the source code (C++), where each source file must contain the name of the main group member developer – **one author per file is allowed**; you should check that **the code compiles on the Virtual Lab**, that is considered the official building environment;
- A report (maximum two pages of text excluding images and tables) presenting your approach and **the performance measurement on the dataset provided and linked above**. You shall report **the metrics and the output images** for every element in the dataset.

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[1] The provided dataset has been extracted from the publicly available DAVIS and SegTrack2 datasets.
https://davischallenge.org/davis2016/soa_compare.html
https://web.engr.oregonstate.edu/~lif/SegTrack2/dataset.html.

[2] https://pyimagesearch.com/2016/11/07/intersection-over-union-iou-for-object-detection/